

Tuition Class Management System

Role-Wise Actions, System Responses, Access Control & Real-Time Notification Specification

DOCUMENT VERSION

v2.4.0 (Enterprise Specification)

RELEASE DATE

September 2026

SYSTEM ARCHITECTURE

React + Express + Prisma (PostgreSQL / SQLite)

DOCUMENT CLASSIFICATION

Technical & Operational Standard (RBAC)

Table of Contents

1. Document Purpose & System Overview	Section 1	2. Role Hierarchy & Personas	Section 2
3. Action Status & Response Taxonomy	Section 3	4. Admin Action Matrix	Section 4
5. Teacher / Faculty Action Matrix	Section 5	6. Student Action Matrix	Section 6
7. Accountant Action Matrix	Section 7	8. Parent / Guardian Monitoring Matrix	Section 8
9. Master Access Control Matrix (RBAC)	Section 9	10. Action Execution & Lifecycle Flows	Section 10
11. Real-Time SSE Notification Matrix	Section 11	12. Standardized API Response Matrix	Section 12
13. Core UX & Security Principles	Section 13	14. End-to-End Action Flow Architecture	Section 14
15. Audit & Activity Logging Specification	Section 15	16. Executive Role Summary	Section 16

1. Document Purpose & System Overview

This technical specification document defines the complete functional blueprint, role-based access controls (RBAC), user action parameters, system responses, real-time notification dispatches, and security policies for the **Tuition Class Management System** (Allen / Aakash / PW enterprise architecture).

The system orchestrates educational and financial operations across five core user roles: **Administrator**, **Teacher (Faculty)**, **Student**, **Accountant (Fee Desk)**, and **Parent / Guardian**. Every interactive module is backed by rigorous backend JWT validation, role-based authorization middleware, and automated real-time Server-Sent Events (SSE).

2. Role Hierarchy & Personas

Role Key	User Persona	Primary Responsibilities	Access Scope
ADMINISTRATOR	Institute Director / Super Admin	Full institute governance: student admissions, faculty hiring, course/batch creation, fee policies, global announcements, and institutional reporting.	FULL SYSTEM (GLOBAL)
TEACHER	Subject Specialist / Faculty	Classroom teaching: attendance tracking, homework assignment creation/grading, study material uploads, doubt clearing, and leave approvals.	ASSIGNED BATCHES ONLY
STUDENT	Enrolled Candidate	Active learning: viewing schedules, submitting assignments, downloading materials, asking doubts, applying for leaves, and viewing fee receipts.	OWN ENROLLED PROFILE ONLY
ACCOUNTANT	Financial Desk / Cashier	Fee management: assigning fee plans, collecting installments (Cash/UPI/Cheque/Bank), generating GST receipts, and auditing revenue.	FINANCIAL & STUDENT LEDGER
PARENT	Guardian / Parent	Parental oversight: read-only monitoring of student attendance percentage, exam ranks/scores, fee payment status, and institute circulars.	LINKED CHILD PROFILE (READ-ONLY)

3. Action Status & Response Taxonomy

AVAILABLE

 Feature actionable by user

RESTRICTED

 Visible but locked

ACCESS DENIED

 Blocked at UI & API

SUCCESS

 Action successfully executed

FAILED

 Execution or validation error

PENDING

 Awaiting approval/grading

NOT APPLICABLE

 Irrelevant to role

4. Administrator Action Matrix

The Administrator holds unrestricted governance over academic, financial, faculty, and system configuration modules.

#	Module / Page	Action	User Input	Permission	Status	System Response	Notification	Expected Result
1.1	Dashboard	View Overview Stats	None (Auto-load)	ADMIN	AVAILABLE	200 OK: Returns student counts, faculty workload, revenue, attendance %.	None	Dashboard metric cards & charts rendered.
1.2	Students	Register Student	Name, Email, Phone, Course, Batch, Avatar	ADMIN	AVAILABLE	201 Created: "Student registered successfully".	Account details sent to Student	Student record created with unique ID; User login created.
1.3	Students	Edit / Deactivate	Updated fields / Status toggle	ADMIN	AVAILABLE	200 OK: "Student profile updated".	None / Status alert	Database record modified; UI table reloaded.
1.4	Faculty	Add Faculty Member	Name, Subject, Specialization, Phone, Email	ADMIN	AVAILABLE	201 Created: "Faculty profile created".	Credentials dispatched	Faculty record created; teacher login activated.
1.5	Courses	Create Course	Course Name, Target Exam, Grade, Fee, Duration	ADMIN	AVAILABLE	201 Created: "Course added successfully".	None	New course listed for batch allocation.
1.6	Batches	Create & Allocate Batch	Batch Name, Course, Faculty, Classroom, Timing	ADMIN	AVAILABLE	201 Created: "Batch created successfully".	Realtime SSE to Faculty	Batch scheduled; timetable updated across system.
1.7	Attendance	Review & Override	Batch, Date, Student Attendance Statuses	ADMIN	AVAILABLE	200 OK: "Attendance saved successfully".	SSE to Absent Students	Attendance records upserted in bulk; stats updated.
1.8	Fee Desk	Assign Course Fee	Student ID, Total Amount, Discount, Plan	ADMIN	AVAILABLE	200 OK: "Fee structure assigned".	SSE: Fee Structure Created	Student fee ledger initialized with installment schedule.
1.9	Circulars	Broadcast Notice	Title, Message, Type, Target Audience, Files	ADMIN	AVAILABLE	201 Created: "Announcement broadcasted".	Realtime SSE to Audience	Notice card appears in Notification Center for target roles.
1.10	Settings	Update Institute Meta	Institute Name, Address, Tax %, Academic Year	ADMIN	AVAILABLE	200 OK: "Settings saved successfully".	Global settings broadcast	Branding, invoice headers, and tax rates updated.
1.11	Reports	Export Audit CSVs	Report Type (Revenue, Strength, Attendance)	ADMIN	AVAILABLE	200 OK: File stream (CSV attachment).	None	CSV report downloaded to local device.

5. Teacher / Faculty Action Matrix

Teachers manage classroom academic activities, evaluations, doubts, and leave authorizations for their assigned batches.

#	Module / Page	Action	User Input	Permission	Status	System Response	Notification	Expected Result
2.1	Dashboard	Faculty Overview	None (Auto-load)	TEACHER	AVAILABLE	200 OK: Returns assigned batches, pending doubts, grading tasks.	None	Faculty dashboard rendered with daily lecture schedule.
2.2	Batches	View Assigned Batches	Batch Selection	TEACHER	AVAILABLE	200 OK: Batch student roster and timetable slots.	None	Displays batch students, attendance history, and course syllabus.
2.3	Attendance	Mark Batch Attendance	Date, Batch, Student Attendance Statuses	TEACHER	AVAILABLE	200 OK: "Attendance marked successfully".	SSE: Attendance Alert to Absent Students	Attendance stored in DB; student attendance % updated.
2.4	Assignments	Create Homework	Title, Batch, Subject, Due Date, Marks, Files	TEACHER	AVAILABLE	201 Created: "Assignment created".	SSE to Batch Students: New Assignment	Assignment card displayed in student learning portals.
2.5	Assignments	Grade Submission	Submission ID, Marks Obtained, Feedback	TEACHER	AVAILABLE	200 OK: "Submission graded successfully".	SSE to Student: Assignment Graded	Submission marked as GRADED; student views marks & feedback.
2.6	Materials	Upload DPP / Notes	Title, Subject, Chapter, Material Type, Files	TEACHER	AVAILABLE	201 Created: "Study material published".	SSE: New Material Added	Files available for download by enrolled students.
2.7	Doubts	Answer Student Doubt	Doubt ID, Solution Text, Solution Attachments	TEACHER	AVAILABLE	200 OK: "Doubt resolved successfully".	SSE to Student: Doubt Answered	Doubt status changes to RESOLVED; solution viewable by student.
2.8	Leaves	Approve / Reject Leave	Leave ID, Status (APPROVED/REJECTED)	TEACHER	AVAILABLE	200 OK: "Leave request updated".	SSE to Student: Leave Decision	Leave record updated; student attendance ledger marked.
2.9	Circulars	Broadcast Notice	Title, Message, Target (Students/Faculty)	TEACHER	AVAILABLE	201 Created: "Notice broadcasted".	SSE to Target Role	Circular displayed to selected audience.
2.10	Fee Desk	Modify Student Fees	None	TEACHER	ACCESS DENIED	403 Forbidden: "Insufficient permissions".	Security warning	Action blocked; fee modification restricted to Admin/Accountant.
2.11	Settings	Modify System Config	None	TEACHER	ACCESS DENIED	403 Forbidden: "Unauthorized route".	Security warning	Settings menu hidden from UI; API access rejected.

6. Student Action Matrix

Students access their personalized learning portal, course materials, assignments, fee ledgers, and doubt resolution forums.

#	Module / Page	Action	User Input	Permission	Status	System Response	Notification	Expected Result
3.1	Dashboard	Student Overview	None (Auto-load)	STUDENT	AVAILABLE	200 OK: Personal attendance %, pending homework, exams.	None	Student dashboard rendered with quick-action tiles.
3.2	My Batch	View Timetable	None	STUDENT	AVAILABLE	200 OK: Daily timetable, room number, faculty details.	None	Displays weekly schedule and upcoming lecture slots.
3.3	Attendance	View Attendance Stats	Month / Subject filters	STUDENT	AVAILABLE	200 OK: Present, Absent, Late logs & overall %.	None	Shows attendance calendar with low-attendance warnings.
3.4	Assignments	Submit Homework	Assignment ID, Solution Files, Comments	STUDENT	AVAILABLE	201 Created: "Assignment submitted".	SSE to Teacher: New Submission	Status changes to SUBMITTED; submission file accessible by faculty.
3.5	Materials	Download DPP / Notes	Material Selection	STUDENT	AVAILABLE	200 OK: File stream / direct media URL download.	Download logged	File downloaded to student device; download count incremented.
3.6	Doubts	Ask Subject Doubt	Subject, Topic, Question Text, Image Attachment	STUDENT	AVAILABLE	201 Created: "Doubt posted to faculty".	SSE to Subject Teacher	Doubt ticket created with status OPEN.
3.7	Leaves	Submit Leave Request	Start Date, End Date, Reason Text	STUDENT	AVAILABLE	201 Created: "Leave applied successfully".	SSE to Faculty & Admin	Leave request recorded with status PENDING.
3.8	Fees	View Fee Ledger	None	STUDENT	AVAILABLE	200 OK: Total fee, Paid amount, Balance, Receipts.	None	Displays official payment receipts with instant download button.
3.9	Circulars	View Announcements	Filter Type / Search Query	STUDENT	AVAILABLE	200 OK: Returns relevant institute announcements.	Unread count badge	Displays circular text, notices, and PDF attachments.
3.10	Other Profiles	View Other Students	Student ID	STUDENT	ACCESS DENIED	403 Forbidden: "Horizontal access denied".	Security log	Student restricted strictly to own profile and results.
3.11	Global Config	Admin Configuration	None	STUDENT	ACCESS DENIED	403 Forbidden: "Route not permitted".	Security log	Admin modules hidden; direct URL access redirects to 403.

7. Accountant / Fee Desk Action Matrix

The Accountant manages fee structures, records cash/online/cheque payments, issues receipts, and conducts audits.

#	Module / Page	Action	User Input	Permission	Status	System Response	Notification	Expected Result
4.1	Dashboard	Finance Overview	None (Auto-load)	ACCOUNTANT	AVAILABLE	200 OK: Revenue collected, pending dues, overdue accounts.	None	Financial dashboard with revenue charts & collection stats.
4.2	Fee Desk	Assign Fee Structure	Student ID, Course, Total Fee, Concession, Plan	ACCOUNTANT	AVAILABLE	200 OK: "Fee structure assigned".	SSE: Fee Structure Created	Student fee ledger initialized with installment schedule.
4.3	Fee Desk	Record Fee Payment	Student ID, Amount, Mode (Cash/UPI/Cheque), Txn Ref	ACCOUNTANT	AVAILABLE	201 Created: "Payment recorded successfully".	SSE to Student: Payment Confirmed	Payment recorded; receipt generated; balance adjusted.
4.4	Receipts	Generate / Print Receipt	Receipt ID / Invoice ID	ACCOUNTANT	AVAILABLE	200 OK: Modal with official branded receipt details.	None	Official receipt displayed with Print & PDF download buttons.
4.5	Audit Register	Filter Dues & Ledger	Date range, Payment mode, Batch filter	ACCOUNTANT	AVAILABLE	200 OK: Filtered financial transactions table.	None	Displays itemized audit log with total collection totals.
4.6	Students	View Student Account	Student Selection	ACCOUNTANT	AVAILABLE	200 OK: Student contact, batch, fee ledger history.	None	Shows complete fee history and remaining balance.
4.7	Circulars	Broadcast Fee Notices	Title, Message, Target (Student/Parent/All)	ACCOUNTANT	AVAILABLE	201 Created: "Notice broadcasted".	SSE to Target Role	Urgent fee reminders broadcasted to selected groups.
4.8	Academics	Create/Edit Courses	None	ACCOUNTANT	ACCESS DENIED	403 Forbidden: "Academic module locked".	Security warning	Accountant cannot modify academic curriculum or batches.
4.9	Attendance	Modify Attendance	None	ACCOUNTANT	ACCESS DENIED	403 Forbidden: "Read-only attendance access".	Security warning	Attendance modification restricted strictly to Faculty/Admin.

8. Parent / Guardian Monitoring Matrix

Parents monitor their child's academic growth, daily attendance, exam performance, homework submissions, and fee status.

#	Module / Page	Action	User Input	Permission	Status	System Response	Notification	Expected Result
5.1	Parent Portal	Child Academic Overview	Linked Child Profile	PARENT	AVAILABLE	200 OK: Summary of attendance %, recent exam ranks, fees.	None	Parent summary view rendered with performance charts.
5.2	Attendance	View Daily Attendance	Month / Date Range	PARENT	AVAILABLE	200 OK: Child's present/absent logs & monthly %.	Low Attendance Alert	Shows real-time attendance calendar and absentees.
5.3	Assignments	Track Homework Status	None	PARENT	AVAILABLE	200 OK: Assigned, submitted, and graded homework.	Homework Pending Alert	Displays submission timeliness, marks obtained, teacher remarks.
5.4	Examinations	View Test Results & Ranks	Exam selection	PARENT	AVAILABLE	200 OK: Scorecard with marks, percentile, All-India Rank.	Result Published Alert	Displays subject-wise analysis and comparative performance.
5.5	Fee Ledger	View Dues & Receipts	None	PARENT	AVAILABLE	200 OK: Fee structure, paid installments, balance due.	Payment Confirmation	Displays official receipts with print and download options.
5.6	Circulars	View Institute Notices	None	PARENT	AVAILABLE	200 OK: Institute holidays, exam dates, parent-teacher meetings.	New Notice Alert	Displays announcements and circular documents.
5.7	Academic Edits	Modify Assignment/Marks	None	PARENT	ACCESS DENIED	403 Forbidden: "Read-only access".	Security warning	Parent portal is strictly read-only for monitoring purposes.
5.8	Other Students	View Other Students	None	PARENT	ACCESS DENIED	403 Forbidden: "Privacy violation".	Security log	Parents are strictly restricted to their own linked ward(s).

9. Master Access Control Matrix (RBAC)

The table below summarizes the comprehensive permission matrix across all 15 operational modules of the system.

System Module	Administrator	Teacher / Faculty	Student	Accountant	Parent / Guardian
Dashboard & KPIs	FULL OVERVIEW	FACULTY KPI	LEARNING KPI	FINANCE KPI	CHILD KPI
Student Profiles	FULL CRUD	VIEW ASSIGNED	VIEW OWN	VIEW ACCOUNTS	VIEW CHILD
Faculty Profiles	FULL CRUD	VIEW OWN	VIEW TEACHERS	NO ACCESS	VIEW TEACHERS
Courses & Curriculum	FULL CRUD	VIEW ONLY	VIEW ENROLLED	VIEW FEES	VIEW ENROLLED
Batches & Timetable	FULL CRUD	ASSIGNED BATCHES	MY BATCH	VIEW BATCHES	CHILD BATCH
Attendance System	FULL CRUD	MARK ASSIGNED	VIEW OWN	VIEW ONLY	VIEW CHILD
Homework Assignments	FULL OVERSIGHT	CREATE & GRADE	SUBMIT & VIEW	NO ACCESS	VIEW STATUS
Study Materials / DPP	FULL OVERSIGHT	UPLOAD/MANAGE	DOWNLOAD	NO ACCESS	VIEW TITLES

System Module	Administrator	Teacher / Faculty	Student	Accountant	Parent / Guardian
Doubt Resolution Forum	FULL OVERSIGHT	ANSWER DOUBTS	ASK DOUBTS	NO ACCESS	VIEW SOLVED
Student Leave Desk	APPROVE/MANAGE	APPROVE/REJECT	APPLY LEAVE	NO ACCESS	VIEW STATUS
Fee Ledger & Receipts	FULL CRUD	NO ACCESS	VIEW OWN DUES	COLLECT & AUDIT	VIEW RECEIPTS
Announcements / Alerts	BROADCAST ALL	BROADCAST CLASS	VIEW NOTICE	BROADCAST FEES	VIEW NOTICE
System Settings	FULL CRUD	NO ACCESS	NO ACCESS	NO ACCESS	NO ACCESS
Audit & CSV Export	FULL EXPORT	NO ACCESS	NO ACCESS	EXPORT FINANCE	NO ACCESS

10. Action Execution & Lifecycle Flows

Every user action follows a rigid, deterministic lifecycle that encompasses frontend submission, payload validation, authorization checks, database transactions, real-time dispatches, and user feedback.

Flow 1: Student Submits Homework Assignment

1

User Action
Student selects assignment, uploads PDF/image files (Max 25MB), enters submission text, and clicks "Submit Homework".

2

Validation & Security Checks
System verifies token authentication, MIME-types, file sizes, and confirms deadline status (calculates if submission is On-Time or Late).

3

Backend Storage & Database Transaction
File stored in persistent storage (/assignments); AssignmentSubmission record upserted with isLate flag and timingText.

4

System Response & Real-Time Notification
Returns 201 Created: "Assignment submitted successfully". Real-time SSE event notification:new dispatched to the batch teacher.

5

Expected UI State
Student card badge changes to "SUBMITTED"; assignment appears instantly in teacher's "Pending Grading" queue.

Flow 2: Accountant Records Fee Payment & Issues Official Receipt

1

User Action
Accountant opens "Record Payment" modal, selects student, inputs amount, payment mode (Cash/UPI/Cheque), and transaction reference.

2

Validation & Ledger Reconciliation
Backend verifies positive amount, validates student ledger existence, and ensures payment does not exceed outstanding balance.

3

Database Transaction & Receipt Generation
Payment record inserted; auto-generates sequential Receipt ID (e.g. REC-2026-0042); student ledger balance decremented.

4

System Response & Notifications
Returns 201 Created with receipt object. Real-time SSE notification sent to student: "Fee payment of ₹X recorded".

5

Expected UI State
Official branded receipt modal opens with print capability; payment appears in audit register and student's fee ledger.

Flow 3: Faculty Marks Batch Attendance

1

User Action
Teacher opens Attendance Sheet, selects batch and date, toggles statuses (Present, Absent, Late), and clicks "Save Attendance".

2

Validation & Processing
Backend validates faculty assignment to batch; bulk upsert executed on Attendance table; attendance percentages recalculated.

3

System Response & Alerts
Returns 200 OK: "Attendance marked successfully". Automated SSE alert dispatched to absent students and parents.

11. Real-Time SSE Notification Matrix

The system utilizes a high-performance Server-Sent Events (SSE) bus (/api/events/stream) with fallback in-app polling to deliver instantaneous alerts across user roles.

Event Key	Triggered By	Recipient	Channel	Message Template / Purpose	Delivery Status
assignment:new	Faculty / Admin	Batch Students	In-App SSE	"New Assignment: '{title}' has been published for {subject}. Due Date: {date}."	IMPLEMENTED
assignment:submit	Student	Faculty / Admin	In-App SSE	"Student {name} submitted homework for '{title}'."	IMPLEMENTED
assignment:graded	Faculty	Student / Parent	In-App SSE	"Your submission for '{title}' was evaluated. Score: {marks}/{total}."	IMPLEMENTED
attendance:marked	Faculty / Admin	Student / Parent	In-App SSE	"Attendance update: You were marked {status} for {subject} on {date}."	IMPLEMENTED
attendance:low	System Monitor	Student / Parent	In-App SSE / SMS	"Urgent: Attendance dropped below 75% ({pct}%). Contact administration."	IMPLEMENTED
payment:recorded	Accountant / Admin	Student / Parent	In-App SSE	"Payment of ₹{amount} received via {mode}. Receipt #{receiptId} generated."	IMPLEMENTED
fee:pending_reminder	Accountant / Admin	Student / Parent	In-App SSE / SMS	"Fee Reminder: An outstanding balance of ₹{balance} is due on {dueDate}."	IMPLEMENTED
doubt:created	Student	Subject Faculty	In-App SSE	"New Doubt: Student {name} posted a doubt in {subject}: '{topic}'."	IMPLEMENTED
doubt:resolved	Faculty	Student	In-App SSE	"Faculty answered your doubt in {subject}: '{topic}'. Click to view solution."	IMPLEMENTED
leave:applied	Student / Faculty	Admin / Faculty	In-App SSE	"Leave Application: {name} applied for leave from {start} to {end}."	IMPLEMENTED
leave:decision	Admin / Faculty	Applicant	In-App SSE	"Your leave application from {start} to {end} has been {status}."	IMPLEMENTED
notification:circular	Admin / Faculty / Acc.	Multi-Role Audience	In-App SSE	"{type}: {title} - {message preview}. Check circulars for documents."	IMPLEMENTED

12. Standardized API Success & Error Responses

HTTP Status Code	Action Status	Standard System Response Message	Triggering Scenario
200 OK	SUCCESS	{ success: true, data: [...], message: "Operation completed" }	Data retrieved, attendance updated, submission graded.
201 Created	SUCCESS	{ success: true, data: {...}, message: "Resource created" }	Student registered, payment recorded, homework created.
400 Bad Request	FAILED	{ success: false, message: "Validation error: [Field] required" }	Missing mandatory input, invalid date range, malformed payload.
401 Unauthorized	ACCESS DENIED	{ success: false, message: "Invalid or expired session token" }	Unauthenticated request, expired JWT token, missing bearer token.
403 Forbidden	ACCESS DENIED	{ success: false, message: "Access Denied: Insufficient role permissions" }	Student attempting fee modification; Teacher accessing Admin config.
404 Not Found	FAILED	{ success: false, message: "Requested record or resource not found" }	Invalid student ID, non-existent batch, deleted assignment.
409 Conflict	FAILED	{ success: false, message: "Duplicate record: Username/Email already exists" }	Duplicate student admission number, duplicate exam ID.
500 Internal Error	FAILED	{ success: false, message: "Internal server error occurred" }	Database connection drop, unhandled server exception.

13. Core UX & Security Rules

Fundamental Security Axiom: "Hidden UI ≠ Security"
Hiding buttons or navigation items on the frontend improves user experience but provides **zero cryptographic security**. The backend API must independently validate authentication and role permissions on every incoming HTTP request.

13.1 Frontend Defensive UX Principles

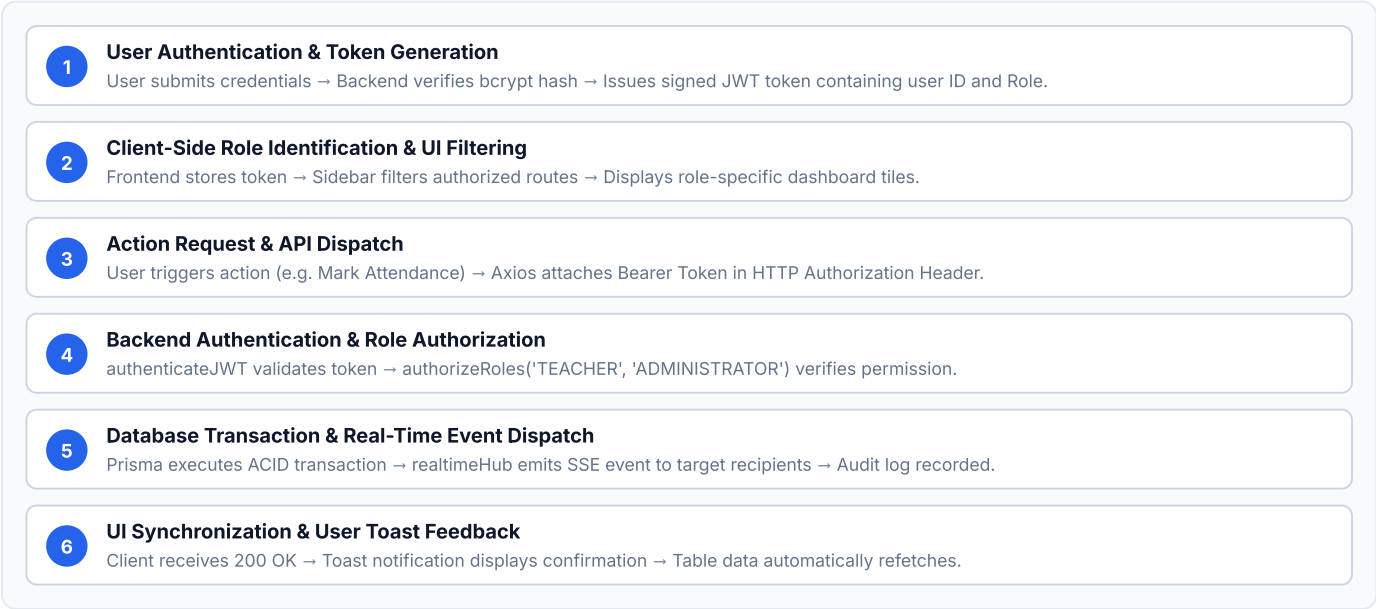
- Role-Based Navigation Stripping:** Navigation links, action buttons, and modal triggers must only be rendered if `user.role` matches the required role. Unauthorized users must never see dead or disabled controls.
- Client-Side Route Guards (`<RoleGuard>`):** Any attempt by a student to manually enter `/settings` or `/faculty` in the browser URL bar immediately triggers the `<AccessDenied>` component and redirects to `/dashboard`.
- Optimistic & Synchronous Feedback:** Action buttons display loading spinners (`isLoading`) during network calls to prevent duplicate submissions.

13.2 Backend Defensive Security Architecture

- JWT Authentication Middleware (`authenticateJWT`):** Verifies token signature, claims, expiration, and user account status (`status === 'ACTIVE'`).
- Role Authorization Middleware (`authorizeRoles(...)`):** Enforces strict RBAC at the controller level before executing business logic.
- Horizontal Privilege Isolation:** When students fetch attendance, homework, or fee receipts, the backend enforces `where: { studentId: req.user.studentId }`, preventing students from inspecting peers' data by modifying URL parameters.

14. End-to-End Action Flow Architecture

The visual sequence below outlines the step-by-step execution path of all authorized interactions:



15. Audit & Activity Logging Specification

Action Category	Actor Role	Target Resource	Severity	Logged Metadata & Context	Retention
AUTH_LOGIN	All Roles	User Session	INFO	IP address, user agent, login timestamp, auth mode.	90 Days

Action Category	Actor Role	Target Resource	Severity	Logged Metadata & Context	Retention
AUTH_FAILED	Anonymous	User Account	WARNING	IP address, target username, failure reason, attempts.	180 Days
STUDENT_CREATE	ADMINISTRATOR	Student / User	INFO	Student ID, Course ID, Batch ID, created by Admin ID.	365 Days
ATTENDANCE_OVERRIDE	ADMIN / TEACHER	Attendance Record	WARNING	Batch ID, Date, Student ID, Previous Status, New Status.	365 Days
PAYMENT_COLLECTED	ACCOUNTANT / ADMIN	Fee Ledger	CRITICAL	Receipt ID, Student ID, Amount, Mode, Txn Ref, Staff ID.	7 Years (Statutory)
ASSIGNMENT_GRADE	TEACHER	Submission	INFO	Submission ID, Assignment ID, Student ID, Score, Evaluator ID.	365 Days
ACCESS_DENIED	All Roles	Secured Endpoint	HIGH	User ID, Role, Attempted URL, HTTP Method, Source IP.	365 Days

16. Final Role-Wise Executive Summary

1. Administrator

Scope: Global Institutional Governance

Unrestricted authority over student admissions, faculty employment, course curriculum, batch schedules, financial fee policies, institutional circulars, settings, and CSV audit exports.

2. Teacher (Faculty)

Scope: Classroom & Academic Operations

Responsible for assigned batch rosters, daily attendance marking, homework creation & scorecard evaluation, DPP study material uploads, doubt resolution, and leave approvals.

3. Student

Scope: Personal Learning & Assessment Portal

Access to weekly timetable, attendance records, homework submissions with file uploads, study material downloads, interactive doubt asking, leave requests, and fee payment receipts.

4. Accountant

Scope: Financial Core & Student Ledger Desk

Dedicated access to student fee assignment, cash/UPI/cheque payment recording, instant branded receipt generation, fee dues registers, revenue audits, and fee circular broadcasts.

5. Parent / Guardian

Scope: Read-Only Ward Oversight

Secure monitoring of their enrolled ward's daily attendance records, exam results and All-India ranks, homework submission status, teacher feedback, fee payment history, and institute notices.

Document Approval & Compliance Certification

This specification document serves as the official compliance and technical baseline for the Tuition Class Management System v2.4.0. All future enhancements, mobile apps, and third-party integrations must strictly conform to the role matrices and access control specifications established herein.